

Predicting any observed quantity of the Earth: a design for a global ocean–land–atmosphere model that reads the observational record through a learnable space–time cone

blauewelt/earth — design paper

18 September 2026

Abstract

We specify a two-stage model whose task is to predict any observed quantity of the Earth system — a grid cell, a station, a float profile, a satellite pixel — at any place, time and footprint, from the whole observational record of the ocean, the land and the atmosphere. The design rests on one geometric fact: a quantity at a place and time depends on a cone in space–time whose reach grows with the speed of the fastest signal that carries information about it and is cut off by that signal’s memory, and it influences the mirror image of that cone into the future. Both stages sample this hourglass. Stage 1 reads raw values over the last thirty days through a waist of several cells at the present and writes one embedding per cell and pentad; stage 2 reads those embeddings over two years at log-spaced lags and writes the embeddings of the next year as a residual on a contractive linear propagator. Every value enters through one token schema — value, observed flag, offset from the anchor, footprint, source — so that dense grids at 0.25° , kilometre-scale fields, tile catalogues and point observations are read by the same encoder without resampling to a common grid, and every prediction leaves through one query-based decoder that takes the same coordinates and footprint, which is what makes “predict anything” a mechanism rather than a slogan. The cone’s shape — a drift toward the source, a waist ellipse and a growth ellipse per 1° cell and channel group, a scale and a memory per channel — is learnable under guards that prevent the model from moving its own questions to easy water. We state the data as the model sees it, the geometry, both stages, the training ladder, the evaluation protocol with its null ladder, and the comparisons that decide each design choice, with their verdict rules written before any run. A results section is reserved with the tables it will hold.

1 The problem and the cone argument

The object. The model is a predictor of the observed state of the Earth system, not of one index. Given the record of observations up to a time t , it is asked for the value an instrument would read at a place $\mathbf{x}$, a time $t + \ell$, a depth or height, and a footprint — the area and time span the reading averages over — for any channel the record has ever contained. Transport through a mooring section, an El Niño index, the chlorophyll of a coastal pixel, the temperature at a weather station and the sea level at a tide gauge are all instances of that one query, and the paper’s claim is that a single representation, read by a single decoder, can answer all of them. The Atlantic overturning, the ocean carbon sink, El Niño and sea-surface temperature are the four read-outs the programme tracks because each has a long, continuous, physically meaningful truth series against which a forecast can be scored; they are properties of the instruments, not of the target.

The cone. A driver moving at speed v can reach $\mathbf{x}$ at time t from anywhere within $v\Delta t$ at time $t - \Delta t$. The set of space–time points that can bear on the value at $(\mathbf{x}, t)$ is therefore a cone opening into the past, and its reach at lag Δt is set by the fastest mechanism that carries information about that channel: an equatorial Kelvin wave or the Somali Current for the ocean’s currents and sea level, the atmosphere for wind stress and for the surface fields the atmosphere stirs, nothing that advects for the soil. The cone is also truncated in time, because a channel forgets: wind stress decorrelates in days, sea-surface temperature anomalies persist for months, the deep western boundary current carries a water mass for years. So each channel has a *family* (v, τ) — a speed and a memory — and its cone is a different shape from its neighbour’s in the same token. Run forward, the same argument gives the set of points that the value at $(\mathbf{x}, t)$ can bear on: the point mirror of the past cone through the anchor.

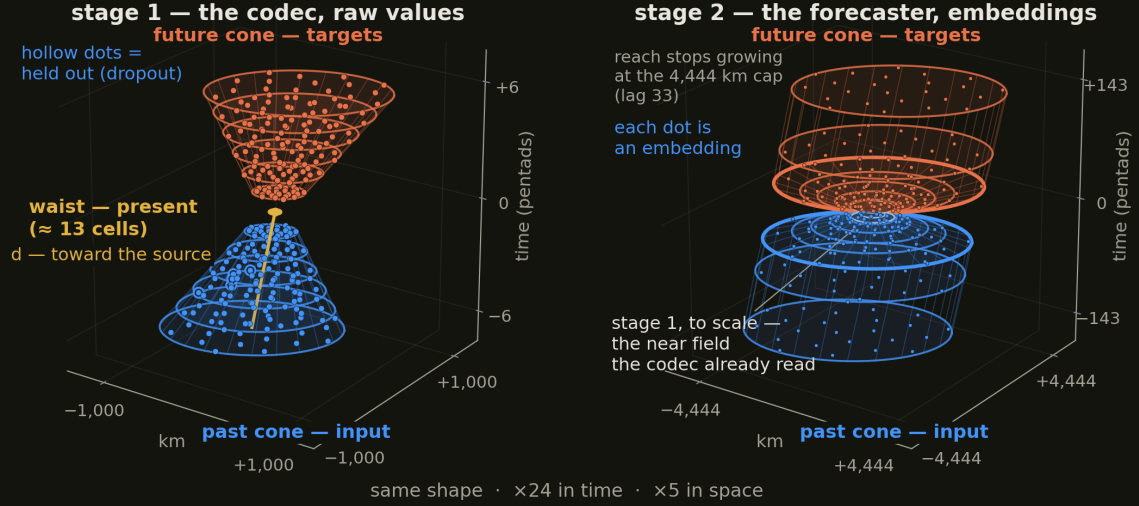


Figure 1: The hourglass at both scales. Each lag is a footprint of 24 sampling points (a Vogel sunflower, phase-rotated by the golden angle per lag so consecutive slices interleave) inside an ellipse whose size grows with lag; the past half is input, the waist is the present, the future half is target only. Left: stage 1 over raw values, six pentads each way, ≈ 900 km at six pentads for the ocean-surface channels, a waist of thirteen cells. Right: stage 2 over stage-1 embeddings, 7–143 pentads each way, opening to a 4,444 km cap; stage 1’s whole hourglass is the small solid at the centre.

A quantity’s past cone is what a model should read; its future cone is what a model should be asked to predict; and the two together are an hourglass with the present at its waist (Figure 1). Advection is what the mirror looks like in space–time: water that reaches the anchor came from upstream and will go downstream, so a cone that leans upstream in the past leans downstream in the future.

Two scales, one shape. The record at 0.25° and five-day cadence is of order 10^9 values per year; at the kilometre scale it is three orders of magnitude more, and the point observations are irregular in every coordinate. No transformer attends over that directly. The design therefore compresses in two stages that sample the same hourglass at different scales. Stage 1, the codec, reads raw values inside a small cone — the last thirty days at full cadence, a few hundred to a few thousand kilometres — and writes one embedding per cell and pentad; its job is to make the near field *attendant*, and it is the only stage that ever sees a raw value. Stage 2, the forecaster, reads those embeddings inside a large cone — thirty-five days to two years, up to the planetary scale — and writes the embeddings of the future cone; it is the stage that is rolled forward and the stage every read-out is scored on. Stage 1’s whole hourglass is the small solid at the centre of stage 2’s. The division of labour is a division in time, not in distance.

Premises. Four premises shape the choices below; they are design decisions, stated once. (i) At the data volume of one basin at pentad cadence the forecasting problem is data-limited: a learned forecaster’s held-out optimum arrives early and widening the model does not move it, so the design grows data — the globe, the other spheres, finer footprints, point observations — before it grows parameters, and every rung of the training ladder is bought on the rung below it. (ii) A linear inverse model in the embedding space and a damped-persistence null are strong forecasts of an anomaly field at leads beyond two weeks, so the learned forecaster is a residual on a contractive linear propagator, not a replacement for it, and is reported against both. (iii) A read-out with a few effective degrees of freedom — a twenty-year transport series, a decadal index — cannot resolve a design choice; comparisons are decided on the dense field and the read-outs are downstream checks. (iv) The holdout is terminal and spatial, and the exclusion of held-out data from every window the forward pass can see is certified before training rather than assumed. A preceding technical report on the North Atlantic pilot of this programme [1] is the dated record of the measurements behind these premises; this paper states the design on its own terms and does not repeat them.

2 The data as the model sees it

Nothing is resampled to a common grid at storage time. Every source is stored at its own resolution and cadence in one of three tiers, the sampler turns all of them into one token at read time, and the decoder is queried at a location *and* a footprint. The programme’s families of data are the instances of these tiers; this section describes them as the model sees them.

2.1 The global tensor

Grid and time axis. The dense backbone is a global tensor on a point-aligned 0.25° grid — latitudes $-90 + 0.25i$, $i = 0, \dots, 720$, longitudes $-180 + 0.25j$, $j = 0, \dots, 1439$, so $721 \times 1,440 = 1,038,240$ cells, rows south first, longitude wrapping at the dateline — at a fixed five-day bin, the *pentad*, counted from 1982-01-01: bin b covers days $[5b, 5b + 5)$, and the record 1982-01 to 2024-12 holds 3,142 bins. A bin is an averaging window, not a date; a year holds 73 of them and none aligns with a month. A coarse 1° grid of 181×360 points is defined the same way, and a coarse channel is read at a fine position by nearest coarse point, so the sixteen fine cells under one coarse cell see one number. That is the model’s honest view of a coarse product and it is not interpolated.

Channel groups. Channels are stored in groups, each at its own native grid and cadence, each an array $[T, H, W, C]$ in float16, bin-major so that one time step of one group is one contiguous byte range:

- **ocean surface** at 0.25° , every pentad: surface current speed and its two components, sea-surface height and log mixed-layer depth from the GLORYS reanalysis (1993–), sea-surface temperature and sea-ice concentration from OISST (1982–) — seven channels, defined over ocean cells;
- **atmosphere and land** at 1° , every pentad, over every surface: wind stress and its within-pentad standard deviation (four channels), 2 m air temperature, 10 m wind (two), surface pressure, precipitation rate, snow water equivalent, soil moisture and temperature in the top 10 cm, the latent and sensible heat fluxes, and the skin temperature — fifteen channels from the NCEP reanalysis, with ERA5 at the same grid as the intended replacement;
- **ocean interior** at 1° and monthly cadence: temperature and salinity at sixteen pressure levels from 10 to 1,900 dbar from the Roemmich–Gilson Argo climatology (2004–), 32 channels, each month written into the pentad that contains its fifteenth day;
- **ocean colour** at 0.25° from September 1997: the block mean of $\log_{10}$ chlorophyll over the 6×6 OC-CCI 4 km cells under each point and the fraction of cell-days in the block that were observed — the second channel says how much evidence the first rests on;
- **statics**: a surface class per cell (702,642 ocean, 226,495 land, 107,074 ice sheet or glacier, 2,029 inland water; ice takes precedence over ocean on an ice shelf because that is the surface an instrument sees) and elevation.

No channel is masked by sphere. A land cell reads its skin and air temperature and its soil state, an ocean cell its surface fields, and every cell reads the shared atmosphere; the observed sea-surface and land-surface temperatures are separate channels because they measure different things with different instruments. A channel is a quantity and an instrument, never a sphere.

Standardisation, transforms and missing values. Every stored value is standardised, $x' = (x - \mu)/\sigma$, with (μ, σ) over all finite values of the channel. Anomalies are not stored: at load time every channel enters as a standardised anomaly against a day-of-year harmonic climatology (annual and semi-annual terms) fitted on training years only and divided by a training-year standard deviation, so nothing in the climatology, the scale or the normalisation of any read-out sees a held-out year. Log-normal quantities (mixed-layer depth, chlorophyll, precipitation, snow water equivalent) are transformed before standardisation; bounded fractions (sea-ice concentration, soil moisture) through a logit. A value that does not exist at a cell and bin — a land cell’s sea-surface temperature, a month without an Argo field, a colour cell under cloud — is not a zero: every token carries an *observed* flag per channel, an unobserved entry is masked from the loss and from the encoder’s input, and the tensor’s own holes are the sources’ holes, counted per channel and bin at build time.

2.2 The fine-granularity families

The 0.25° pentad tensor is the coarsest thing the model reads. Three facts decide how finer sources enter. A coarse value copied into fine cells is a lie the model cannot see through: nine copies of one reanalysis value are not nine measurements. A fine value averaged into a coarse cell throws away exactly what the finer source was bought for: a 4 km colour field averaged into 0.25° keeps one part in thirty-six. And *missing* is not damage and neither is *far away*: the nearest profile to a cell is information whose value depends on its distance, and a grid cell hides that distance. So finer sources are stored by their density — what a read costs — rather than by their sphere or their science, in three tiers with one registry:

- **Tier G, gridded dense:** fields at or finer than about 4 km, daily to monthly, near-complete coverage — ocean colour at 4 km (OC-CCI, 1997–), level-3 sea-surface temperature at 5.6 km per sensor (SST CCI) and 2.2 km merged (ACSP0), sea ice at 6.25 km on polar grids, 4 km geostationary cloud-top temperature, 0.1° daily precipitation (IMERG), and on land the daily 0.05° land-surface temperature, snow and surface reflectance fields. A group finer than 0.25° is stored *sharded*: one compressed file per group and pentad, holding one frame per day (or per three hours), cut into 256×256 -pixel tiles with a byte index per tile, so a read fetches only the tiles around the anchor and an empty tile costs nothing. The footprint is a constant per group.
- **Tier P, points, profiles and tracks:** everything sparse in space or time — Argo and BGC-Argo profiles, every non-Argo ocean profile since 1772 (WOD), ship and buoy reports since 1662 (ICOADS), ship bottle chemistry (GLODAP), surface drifters, the tropical moored arrays, open-ocean moorings, the Atlantic overturning arrays’ own instruments, surface CO_2 (SOCAT), along-track altimetry at 7 km and SWOT swath samples at 2 km, column CO_2 soundings, wave height along track; on land, weather stations and balloon soundings, tide gauges, coastal buoys, gliders, flux towers, active-fire detections, river discharge, soil probes, orbital LiDAR footprints. A store is a set of column arrays — time in integer seconds, latitude, longitude, values, observed mask, quality, a hashed platform identity, a per-row footprint — sorted by pentad bin with an index of where each bin starts, so the k nearest observations to an anchor within a space-time radius are one bounded scan. Observations before 1982 keep negative bins; the reader clips to the axis in use.
- **Tier T, tiles:** fine imagery at or under a kilometre — Sentinel-1 and -2, Landsat, NISAR, OLCI and SLSTR, VIIRS, MODIS at 1 km — is *not* stored as values. The store is a catalogue of scene-dates (footprint polygon, sensor, time, cloud and valid fraction, location) and a *token cache*: a small local codec (§2.3) encodes one tile-date into a few 64-dimensional vectors, which are stored as a tier-P store whose “values” are those vectors, with footprint equal to the tile and the day. Google’s published AlphaEarth embedding [2] — 64 numbers per 10 m of land per year — enters through the same path as a ready-made token cache, which is what allows the programme’s own tile codec to be measured against it with everything else held fixed.

Three properties make the tiers one family. One time axis: every tier is indexed by the pentad bin from the 1982 epoch, tier-P rows keeping their exact time inside the bin so Δt is fractional. One registry: a JSON file per family lists every group with its tier, layout, cadence, channels and units, footprint constants, first bin, file checksums, sources, licence and builder commit, and a consumer dispatches on the tier, so a new group is an entry and not a code path. One sampler contract: `gather(anchor, family)` returns, for each group, tokens in the schema of §2.4 under that group’s cone family, with a fixed slot count per group so that batches stay rectangular. Datasets whose licences forbid redistribution are stored on a private track in the same format, listed in the public registry, and never republished; a model trained on them is published only where the licence’s terms on derived works allow it.

2.3 The local codec for tiles

A 64×64 -pixel tile at 10 m is 4,096 pixels of 13 bands, and a 0.25° cell holds some seven thousand of them; they cannot be sampling points. The local codec is a masked autoencoder over one tile-date — all bands, the sensor’s own metadata (resolution, incidence angle for radar, sun angle for optical) as conditioning, so that one codec with a learned *source* embedding serves optical and radar sensors alike — whose objective is masked-pixel reconstruction plus a same-place-different-date contrastive term, so

that the token carries state rather than texture. It outputs $k = 4\text{--}8$ tokens of 64 dimensions per tile-date, cached once and reused by every anchor whose cone contains the tile. It has snapshot semantics by construction: it sees one date, and the cone codec supplies the time axis by reading the cached tokens of many dates. The tile size is a knob to be measured, not a constant: at 640 m tiles and seventy clear dates a year the global token cache is of order 20 TB per year; at 2.56 km it is sixteen times smaller and still a hundred times finer in area than a 0.25° cell. Raw imagery is streamed once to train and once to encode, and never kept.

2.4 One token schema

The unit the encoder attends over is a place at a time, carrying every channel present there:

$$\text{token} = (\mathbf{v}[C], \mathbf{m}[C], \Delta x, \Delta y, \Delta z, \Delta t, \log_2 f_p, \log_2 f_t, n_R, s, \text{sphere}, \text{season}), \quad (1)$$

where $\mathbf{v}$ is the vector of standardised anomalies of the group’s channels and $\mathbf{m}$ the observed flag per channel; $(\Delta x, \Delta y, \Delta z, \Delta t)$ is the offset from the anchor in kilometres, metres of depth or height, and days; $\log_2 f_p = \log_2(\text{footprint km}/27.83)$ and $\log_2 f_t = \log_2(\text{support days}/5)$ are the spatial and temporal support of the measurement in units of a 0.25° cell and a pentad, clamped to ± 12 (a Sentinel-2 pixel reads $(-11.4, -4)$, an Argo profile $(-4, -4)$, a 4 km daily colour cell $(-2.8, -2.3)$, an OISST pentad cell $(0, 0)$, a reanalysis cell at its true spectral support $(+2.9, 0)$, a monthly 1° Argo field $(+2, +2.6)$); n_R is how many observations of this kind were within reach; s is a learned source embedding (sensor or product class); and the sphere code and the day-of-year harmonics of the token’s own time complete it. One token per sampled location and lag, *not* one per channel per location: the channel dimension lives inside the token, so the token count is the geometry’s and not the channel count’s, and adding a channel costs a vector entry rather than a token. The two footprint fields are constants per group for dense sources and per row for sparse ones, and nothing else in the schema depends on the tier: the encoder never learns which file a value was read from, only what it is, where and when it is, and how much it averaged.

Offsets are what correct the coarse-value problem. A 1° cell’s value is one token located at the cell centre, seen from sixteen different $(\Delta x, \Delta y)$ by the sixteen anchors inside it, and the present-day patch of a fine anchor holds one such token, not nine copies. Point observations enter as tokens of the same schema placed where and when they were observed — the k nearest profiles, drifters or stations in space–time within the cone, k small (three to eight per depth group), one-sided in time, padded to a fixed k with *miss* tokens — and there is no ellipse to learn for them, only the attention over what was measured. A token whose value the encoder is not allowed to see carries a *hidden* marker, distinct from the *never observed* marker: a value we hid and a value the data never had are different facts.

2.5 The holdout

Three development years — 2009, 2017 and 2023 — are held out of training, of the climatology and of every statistic, for model selection; the terminal test is one scoring of a model trained on data up to 2020 on 2021–2024, run once. A training example is admissible only if every bin its forward pass can see — every past-cone token, every future-cone target — is a training bin, and that admissibility is certified by brute force over a large draw of anchors before a run starts; a non-zero count refuses the run. Longitude holes are held out in addition, so that extrapolation into an unseen region is scored separately from interpolation inside a seen one.

3 The hourglass and its learnable geometry

3.1 Waist, past cone, prediction cone

The waist. The present is a disc of radius two cells at 0.25° — about thirteen cells, sampled as a small sunflower so that every bearing is represented — read as input and reconstructed, and the embedding z is written for the anchor at its centre. The waist exists because the quantities the read-outs depend on are gradients: thermal wind is a horizontal density gradient, geostrophic velocity a slope in sea-surface height, and a gradient does not exist inside one cell. The waist carries the same shape parameters as the cone evaluated at lag zero, so a boundary current can have a waist elongated along the flow.

The past cone. Lags 1–6 pentads, every pentad, so that the codec owns the last thirty days at full cadence: five days is the finest cadence the dense tensor has, and the thirty-day window is the codec’s whole job. At each lag and for each channel group the dense sampling set is a 24-point sunflower — the Vogel spiral at the golden angle — mapped through an ellipse. The footprint at lag ℓ is Gaussian with centre $\ell \mathbf{d}$ from the anchor and covariance

$$\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v, \quad p(\ell, i) = \mathbf{x}_{\text{anchor}} + \ell \mathbf{d} + \Sigma(\ell)^{1/2} \mathbf{u}_i, \quad (2)$$

where $\mathbf{u}_i$ are the canonical sunflower points on the unit disc, $\mathbf{d}$ is the drift per pentad — the arrow toward the source — Σ_0 is the waist ellipse and Σ_v the velocity-spread ellipse, so that the footprint’s standard deviation grows linearly with lag, which is the reach-grows-with-lag rule of the cone argument. Time is not sunflower-sampled — one slice per pentad — but a stack of identical slices repeats the same bearings at every lag, so the sunflower’s phase advances by the golden angle from one lag to the next (bearing $_{i,\ell} = (i + \ell) \times 137.5^\circ$) and the six slices interleave in space–time at no cost. The anchor’s own column is always sampled at every lag.

The far ring. In addition to the ellipse, every lag keeps one fixed ring of 24 points at log-spaced radii from 28 km out to the lag’s design reach $v_{\text{design}} \Delta t (1 + \ell)$, capped at the antipode, read at full weight and never gated. The design speed is the maximum measured speed of the fastest mechanism that carries the family’s information, times a safety factor of 1.5 (for the ocean, the Somali Current and the first-baroclinic Kelvin wave at 3.6 m s^{-1} , so 5.4 m s^{-1} and 2,333 km per pentad; for wind stress the atmosphere, which is global within one pentad). The factor covers three things at once: a pentad mean on a 0.25° grid under-reads peaks, the maxima are literature values rather than the true extreme, and a warming ocean may run faster. The ellipse gives the cone direction and density; the ring guarantees that no driver is excluded by construction. A cone that is too wide costs tokens, a cone that is too narrow excludes true causes silently, and the buffer is spent on the cheap side.

The prediction cone. The past cone point-mirrored through the anchor: centre $-\ell \mathbf{d}$, the same $\Sigma(\ell)$, leads +1 to +6 pentads, sampled the same way and read by nobody. Each point in it is a decoder query and a target only. The anchor column at every lead is always among the targets, which is what makes the fixed-frame forecast that stage 2 consumes always trained.

3.2 Channel families and the per-channel aperture

One direction per flow, one scale and one memory per channel. What is shared per region, lag and channel group is the shape $\Sigma(\ell)$ and the drift $\mathbf{d}$: everything the water carries is carried the same way, so the arrow toward the source and the elongation along the jet are properties of the flow and not of the tracer. What is per channel is a log-scale s_c (two for the L-shaped channels, lags 0–1 and 2–6) and a memory τ_c : in space $\Sigma_c(\ell) = e^{s_c} \Sigma(\ell)$ — the same ellipse inflated or shrunk, never turned — and in time lag ℓ is weighted by $e^{-\ell \Delta t / \tau_c}$, how far back the channel is worth reading. Both are initialised from a table of families set by physics: wind stress at 500 km within a ten-day memory (so lags 0–1 count and lag 6 is at e^{-3}); currents, sea-surface height and the profile tokens at 130 km per pentad with a long memory; sea-surface temperature and mixed-layer depth at max(that, 500 km) at lags ≤ 1 , where the atmosphere stirs them, and the ocean’s growth thereafter; land at 400 km flat and a long memory, because soil does not advect; the depth column with no ellipse of its own. At step zero every channel reads exactly the reach and the lags the table gives it.

The aperture, at zero extra tokens. Because a token carries all channels of its group, each channel reads the group’s shared points through its own weight,

$$w_{c,i,\ell} = \exp\left(-\frac{1}{2} p_i^\top \Sigma_c(\ell)^{-1} p_i\right) \exp\left(-\ell \Delta t / \tau_c\right), \quad (3)$$

applied to that channel’s value and observed-flag entries before the token’s projection. A point outside channel c ’s aperture in space or beyond its memory in time contributes about zero for c and still carries the other channels at full weight; a point 400 km from the anchor at lag 1 is read by the currents at about 30% and by sea-surface temperature at about 73% at the initial apertures, from one token and one read. The aperture is a smooth function of Σ_0 , Σ_v , $\mathbf{d}$, s_c and τ_c , so the loss reaches all of them through every candidate point at once — a far better gradient than the four neighbouring cells an

interpolated read would give, and the reason the geometry can learn without the data loader knowing where it currently is. The far ring is not gated: a safety net the model can switch off is not one.

3.3 The learnable parameters and their guards

Eight dimensionless numbers per 1° cell and channel group. The drift is stored as $\mathbf{d} = \tanh(\mathbf{x}) v_{\text{design}} \Delta t$, capped at the design speed by construction. The two ellipses are stored as matrix logarithms,

$$S = \log \Sigma = m \mathbf{I} + k \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \quad m = \ln \sigma_1 + \ln \sigma_2, \quad k = \ln \sigma_1 - \ln \sigma_2, \quad (4)$$

whose trace is the size and whose traceless part is the log aspect ratio times the double-angle phasor: no parameter is in kilometres, no angle wraps at 180° , a circle is the natural zero, and — the reason for the form — an Adam step of one learning rate moves every radius by the same fraction of itself whatever its size [3], where a radius stored in kilometres would be frozen at a thousand kilometres and twitchy at thirty. The waist’s eigenvalues are floored at one cell (28 km) and Σ_v ’s at the design speed by a softplus offset in log space rather than a clip, so the floor never kills the gradient. The per-channel s_c and τ_c are global, about a hundred numbers in all.

Where they live. On a 1° map per channel group (64,800 cells; ocean maps live over ocean, land maps over land), the anchor’s parameters being the bilinear blend of the four surrounding cells, so the geometry varies smoothly and never jumps at a cell edge. One degree because the currents that carry the overturning are about 100 km wide — the Florida Current runs through an 80 km strait — and a 10° cell would average the Gulf Stream with its recirculation and learn a drift near zero. Against noise in data-poor cells, each map is a coarse 10° map plus a 1° residual under an L2 shrinkage and a Laplacian smoothness prior with one weight, so a cell at the ice edge or along a coast inherits its region’s arrow rather than fitting noise; each 1° cell still sees sixteen anchors over three thousand pentads for its eight numbers. There is one map per channel group — ocean surface; atmosphere, with drift only inside its memory; land, with no drift — and none for the depth column, whose tokens are the nearest profiles with their own offsets and depths. All told about 1.6 M nominal parameters, half of them live, and no compute. A first-harmonic seasonal term on the drift, $\mathbf{d}(\tau) = \mathbf{d}_0 + \mathbf{d}_1 \cos \tau + \mathbf{d}_2 \sin \tau$, is a second-phase option for monsoon regimes where the current reverses.

How they learn: gate first, harden later. In the first phase the candidate points per region and group are fixed — a log-radial set of 36 per lag out to the design reach, plus the anchor column — so that the loader’s reads are regular, cacheable and never depend on live weights, and the geometry learns only through the aperture weights over those candidates. Every N steps (2,000 to start) the dense sunflower table per region is re-baked from the current ellipses and the loader swaps tables at an epoch boundary; within an epoch reads stay regular. The geometry parameters get a learning-rate multiplier of ten over the codec’s, the practice of deformable attention [4], recorded per run. Reading at continuous positions through a deformable gather was considered and rejected: it couples the loader’s read pattern to the live weights, and its gradient sees only the four cells around each point.

Why a learnable cone can cheat, and the guards. If target positions could move, the gradient would move them to smooth open water, away from fronts — a lower loss that means nothing. Five rules close the ways of doing so. The loss reaches the geometry only through the *input* reads; target positions carry a stop-gradient and are treated as constants. The future cone is tied to the past cone by the mirror, so it cannot be tuned on its own; an untied arm is an ablation, not the design. The evaluation geometry is frozen to the fixed hourglass for every arm, so two runs with different learned cones are scored on the same questions. The aperture never enters the loss weights: in the gate phase the targets are the fixed mirrored candidate set with fixed weights, so a model that narrows its aperture gains nothing on what it is scored on. And the far ring is kept and never gated. A residual second-order effect remains — as the input cone moves, the mirrored targets move with it — and is monitored rather than assumed away: every evaluation reports the variance of the targets under the learned cone against the fixed cone, and a drop of more than 10% is a flag that the geometry is drifting toward easy water rather than toward the source.

task	input	targets	share	what it teaches
T1 forecast	waist + past cone	future cone	0.35	the full problem
T2 snapshot forecast	waist only, past cone hidden	future cone	0.15	the built-in twin: what history buys, inside one model
T3 nowcast from history	past cone under a dropout pattern; waist hidden	waist	0.25	the present from the past: persistence, advection, memory
T4 fill-in	waist + past cone, both under a dropout pattern	the hidden waist cells and points	0.25	reconstruction from partial observation

Table 1: The four tasks of stage 1, drawn at random per example. Shares are a starting point and the first thing the comparisons of §8 vary.

Initialisation. $\mathbf{d} = 0$; S_0 the log of an ellipse of the waist’s radius with the measured flow anisotropy of 0.71; Σ_v from the family table’s speed; s_c and τ_c from the same table — the fixed hourglass exactly, so that step zero of the learned arm *is* the control and every departure from it is something the data asked for.

4 Stage 1: the codec

Tasks. Each training example is one anchor at one bin and draws one of four tasks (Table 1). Dropout patterns for the nowcast and the fill-in fire independently, each with its own probability: one channel hidden at the waist only; the most recent lags hidden (every point at lag $\leq \ell_0$, $\ell_0 \in \{1, 2, 3\}$), so the model must extrapolate from stale history; a 90° bearing wedge hidden, so it cannot lean on one direction; points hidden at random ($p = 0.3$), imitating sparse observation; and, for the nowcast only, the whole waist. The task is not told to the model by a token: the hidden and never-observed markers are visible and the lead is in every query’s coordinates, which is enough. Within an example the loss weights are equal across tasks; the shares set the mix and are recorded per run.

Three rules are not knobs. *Knowable targets only*: a target’s channel must be visible somewhere in the input, at some lag, or the target is unknowable and teaches the model to predict the mean; the sampler enforces this and logs the count of violations, which is required to be zero. *No copy targets*: the waist family scores hidden channels only, because a target the model can read off the input is not a question. *Every family against its own bar*: the loss per task, family and lead is reported against the score of predicting the mean, and a family is called learned only where the ratio is below one at held-out anchors.

Model. A masked autoencoder in the strict sense [5]: the encoder sees only visible tokens and visible channel entries, never a placeholder for a hidden one. The encoder is a transformer over the ≈ 300 per-location tokens of one example — Perceiver-style cross-attention into a fixed latent set [6] where a fixed-size interface is wanted, a plain encoder with a learned attention pool otherwise — producing z of width 128–256. The bottleneck is in the token count, not in the width: what the forecaster needs compressed is the near field’s thousands of values into one vector per cell, and a narrow z is where capacity fights between the waist and the cone. The decoder is query-based and reads z alone: a query is (offset, channel, sphere, depth or height, lead, footprint), it cross-attends to z , and its head emits a mean and a log-variance per channel group. A decoder that could also read the encoder’s latents would let z carry nothing; that path is closed. At the ladder’s larger rungs (§7) the feed-forward block becomes a mixture of experts routed by sphere code and channel group, because the value vectors differ by sphere in statistics and physics and expert specialisation is the natural fit at constant active parameters.

Objective. A heteroscedastic Gaussian likelihood per channel group in its transformed space, in the decoupled form

$$\mathcal{L} = \|\mu - y\|^2 + \frac{1}{2} e^{-s} (y - \text{sg}[\mu])^2 + \frac{1}{2} s, \quad s = \log \sigma^2, \quad (5)$$

so that the mean is trained by plain squared error — what the forecaster consumes and what every score measures — while the variance is trained by the likelihood with the mean detached. This removes the known pathology of heteroscedastic regression in which the variance head explains a hard target away instead of the mean learning it [7]; the β -weighted likelihood of that paper at $\beta = 1$ has the same

gradient on the mean. s is clamped to $[-8, 8]$, the first 2,000 steps run at fixed $\sigma = 1$, and the likelihood is per channel group: Gaussian on standardised anomalies for the continuous fields, Gaussian in log space for the log-normal ones, a hurdle model (Bernoulli wet or dry times a log-Gaussian amount) for precipitation, and a logit-Gaussian for bounded fractions. Channel groups are weighted explicitly in the loss as a stated choice, never through σ as a side effect. Every family — waist reconstruction, fill-in, each future lead — is scored at every evaluation against its own predict-the-mean bar; coverage of the $\pm 1\sigma$ and $\pm 2\sigma$ bands on held-out targets is reported as the calibration check.

Training stability. Embedding collapse is invisible to a reconstruction loss, because a decoder rescales freely, so the effective rank of z over a fixed held-out batch and a scale-invariant probe correlation are logged at every evaluation and a run aborts on two consecutive sub-threshold readings. No quantisation is switched on from a random initialisation: a lattice on an untrained encoder collapses within thousands of steps, whereas a lattice added to a finished continuous codec does not, so a discrete code, where wanted, is a second phase. Inputs are standardised, so the gradient clip is 1.0; AdamW with $\beta_2 = 0.95$, weight decay 0.1, a 2,000-step warm-up and cosine decay to 10% of peak; bf16 with fp32 master weights. Training stops at the held-out minimum with the minimum’s step recorded, and no arm runs past four epochs of anchors [8].

5 Stage 2: the forecaster

Inputs. Stage-2 tokens are the codec’s embeddings: at each sampled cell and lag, the mean and log-variance of z , the offset $(\Delta x, \Delta y, \Delta t)$, the statics and the day-of-year harmonics of that bin. The past cone reads lag 0 — which already summarises the last thirty days — and log-spaced lags 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 and 144 pentads, twelve slices instead of 144, each with the anchor column and a 24-point footprint whose reach follows the fastest *slow* signal — planetary waves and gyre adjustment at $0.05\text{--}0.1\text{ ms}^{-1}$, about 1,000 km at 34 pentads, the 4,444 km cap from about lag 33 on — and a far ring to the antipode at every lag, as in stage 1. Dense sampling where predictability is dense and sparse where it is not is what keeps an example at about 300 tokens rather than twenty thousand; the dense 144-pentad window is the ablation. The same eight-number learnable geometry applies at this scale with its own 1° maps and the same guards.

The linear backbone. For the anchor column,

$$\hat{z}_{t+\ell} = \mathbf{G}_\ell z_t + f_\theta(\text{cone}, \ell), \quad \mathbf{G}_\ell = \mathbf{U} \text{diag}(e^{-\ell/\tau_j}) \mathbf{U}^{-1}, \quad \tau_j > 0, \quad (6)$$

where $\mathbf{G}_\ell$ is a linear propagator in embedding space with learned time constants — contractive by construction, so a free run decays toward climatology rather than drifting — initialised from a linear inverse model [11] fitted to the training-year embeddings, and f_θ is the transformer over the cone with a zero-initialised output projection. At step zero the forecaster *is* the embedding-space linear inverse model; whatever it becomes has to earn its distance from it on the held-out field. Damped persistence is the diagonal special case and is reported as its own null.

Direct multi-lead targets. The target is the mirrored future cone: the anchor column and its footprint at leads 1, 2, 3, 5, 8, 13, 21, 34, 55, 73 pentads (73 pentads is one year), each lead a query $(\Delta x, \Delta y, \ell)$ to the same decoder head, predicted directly rather than by chaining one-step predictions. A direct forecast at lead ℓ cannot compound its own errors, its amplitude is set by the data at that lead rather than by an accumulation, and the log spacing puts the targets where the correlation is still changing. The anchor column is weighted 1 and the off-anchor footprint 0.25 (a choice); the footprint targets are what make the prediction a field with coherent advection rather than a set of independent columns. Leads beyond a year are reached by feeding the predicted anchor embeddings at leads 1–73 back as the recent past and querying again: the multi-year roll is a second mode of the same model, not its training objective.

Objective and calibration. The decoupled Gaussian likelihood on each target embedding, normalised per lead by that lead’s predict-the-mean bar so that every lead counts equally and the long, unpredictable leads do not dominate the gradient; plus a decoded-space term through the frozen codec

decoder for the scoreable channels at leads 1, 8 and 34 (squared error against the standardised anomaly, relative to climatology), because the embedding metric is not the metric the forecast is scored on; plus, for the multi-year mode, a two-step pushforward term in which the model is rolled once with a stop-gradient on the first step and scored on the second, so it learns to correct its own errors without back-propagating through the roll. Under a squared-error mean the conditional expectation shrinks toward zero as correlation falls, so amplitude calibration is a property of the objective at the held-out minimum and not a post-hoc fit; the identity $\text{MSSS}_{\text{clim}} = 1 - (1 + a^2 - 2a \text{ACC})$ between the squared-error skill, the anomaly correlation and the amplitude ratio a of a forecast against a zero-anomaly reference is logged per lead as the check that it holds, and a per-lead scaling fitted on training-year starts is reported as a diagnostic only.

Regularisation. Three things keep the forecaster in the regime where it learns rather than memorises. The globe gives every regime of the ocean and atmosphere as training patterns rather than one basin’s; the log-spaced cone reduces the parameters per example by a factor of seventy against a dense window; and the model’s free part is a residual on a linear backbone that the data pin down with few degrees of freedom. On top of these: Gaussian noise on the input embeddings at 0.7 of the persistence scale, a denoising objective in disguise; random dropout of past-cone points; weight decay; early stopping at the held-out minimum with the step recorded; and, because the model is cheap at this token count, a seed ensemble of three to five whose spread is the first probabilistic forecast and whose mean is the deterministic one.

Probabilistic extension. Once the deterministic level exists, a shared low-dimensional noise vector per forecast conditions the head so that samples are coherent across cells and leads, scored by the fair continuous ranked probability score [14]. Explicit categorical marginals over a quantised embedding under the same shared noise are the alternative where the conditional distribution is two-humped, which at six to twenty-four months an El Niño index plausibly is; both are decided on the dense field, never on an index.

6 The queryable read-out: predicting anything

The decoder of §4 is the mechanism by which one model answers every question, and it is worth stating what it can and cannot do.

A prediction is a query. A query is a tuple in the same coordinates as an input token: an offset from the anchor in space, depth or height, and time; a channel; a sphere; a source class; and a footprint ($\log_2 f_p, \log_2 f_t$). “Channel c at point p at lead ℓ over this footprint” is one query, and the same decoder answers a 0.25° cell over a pentad, a 300 m pixel on a day, a station at an hour and a float profile at its depths, because it was trained on targets that carried exactly those coordinates and footprints. The mean chlorophyll of a 0.25° cell over a pentad and the value of a 4 km cell on a day are two different queries of one embedding, and the loss for each is scored against the observation that actually has that support. The footprint on the output side is what lets one model be trained on coarse targets today and on fine targets later without a second head.

Anything the record has contained. The domain of the read-out is the set of channels that appear somewhere in the training targets, at the footprints they appeared with. A channel never observed cannot be asked for; a footprint far outside the range seen is an extrapolation and is reported as such. Between those limits the decoder interpolates in coordinate space, and the same query asked at a place the source never covered — a sea-surface temperature over land — returns a mean with a variance that says so, because the training targets were masked there and the variance head has never been rewarded for confidence.

Indices and sections. A quantity that is a functional of many cells — an overturning transport, an area-mean index, a warm-water volume — is a head on the predicted embeddings of the cells it depends on: an attention head over the individual cells of a section, trained on training-year true embeddings and applied to forecasts, never a head on a spatial mean, because geostrophic transport is the east–west

rung of data	values	tokens	ceiling	what it takes
one basin, 0.25° pentads, 42 ocean channels	2.5 B	0.04 B	0.12 B	the pilot’s tensor
global pentads, four groups and statics	13.8 B	0.21 B	0.7 B	built: the tensor of §2.1
+ the shared atmosphere and land channels of every sphere	91 B	1.4 B	4.5 B	ERA5, radiometers, scatterometers, gravimetry, soil moisture
daily cadence, all spheres	455 B	7.0 B	23 B	a 1.3 TB tensor and a streaming loader
kilometre tier, all spheres	3.5×10^5 B	5.4×10^3 B	—	not a tensor: the token cache of §2.3; compute-bound for the first time

Table 2: The data ladder in observed values; tokens at 65 channels per token; the ceiling at one parameter per twenty values, an upper bound because values are correlated.

contrast across a section and a section mean is the statistic in which that contrast cancels. The read-out is the cheap, replaceable end of the system; none of the read-outs is used to choose between models.

Observations as their own truth. Because point observations are targets in their own coordinates, a store’s own record is the verification of its forecast: a drifter’s velocity, a mooring’s temperature at 300 m, a gauge’s sea level, a tower’s CO₂ flux. Where a store has a licence that forbids redistribution, the score can still be published; the values cannot.

7 Training at scale

The unit of account. Data is priced in observed values and their independence, never in bytes. A 0.25° cell is correlated with its neighbours over a few hundred kilometres and, at the surface, over months, so a basin’s 10^9 values are of order 10^6 – 10^7 independent ones — the regime in which a small model equals a large one. The globe adds new regimes, independent by construction; the other spheres add new physics; daily cadence inside a pentad is mostly the same sample five times; the kilometre tier adds fine-scale variance that is genuinely new. The ladder’s order follows: globe and spheres first, then footprint, cadence last. Table 2 prices the rungs against a Chinchilla-style anchor of one parameter per twenty values [9], quoted as a ceiling because of the correlation above.

The rungs. Each rung of the model is bought only on the previous rung’s read-out, under the frozen protocol, with the null ladder of §8 beside it, and a rung whose held-out minimum arrives inside 2,000 steps at full data is over-parameterised, so the next rung is not bought. L1: a 7–125 M codec on the global pentad tensor with the fixed and learned geometries — the comparisons of §8.1. L2: a 0.3–1 B codec and forecaster on the same tensor plus the tier-P stores, where the width ladder is re-measured, because it is the one measurement that tells a data-limited rung from a compute-limited one. L3: 1–4.5 B with the sphere-routed mixture of experts on all spheres, where the question is whether land and atmosphere improve the ocean’s rolled skill and vice versa. L4: daily cadence, the first compute-bound rung, where the question is whether five times the bytes are worth anything on the terminal holdout. L5: the kilometre tier, only after L4 has a positive answer. Compute is priced at $6NT$ per epoch; a 1 B codec over the global pentad tensor is of order thirty RTX-4090-hours at four epochs, and the encoder’s cost is linear in the token count because the latents cross-attend to the tokens, so the aperture and the far ring cost one elementwise multiply per token and no new reads.

Sampling and throughput. An anchor is a cell and a bin drawn from the admissible set, uniformly over ocean, land and ice in proportion to a stated mix; the gather of one example — the sharded tiles, the point stores’ k -nearest scan, the dense slabs — is host-bound, not accelerator-bound, and is batched by latitude row so that the per-row dot tables are reused, with candidate reads regular within an epoch by the gate-then-harden rule. Every artefact a run produces — the checkpoint, the metrics, the geometry maps — is written incrementally and published to durable storage while the run is alive, so that a job that dies leaves the next one nothing to recompute; a training run’s configuration is a named recipe, never a hand-assembled set of flags, and a resumed run derives its architecture from its

checkpoint. A number from a single seed is a direction, never a level: the first result at every new tier, metric, cadence or codec buys its own replication, and any probe-scored claim needs three seeds.

8 Evaluation protocol and pre-registered comparisons

Protocol. Held-out years 2009, 2017 and 2023 stay held out of training, of the climatology and of every statistic; a codec and forecaster trained on data through 2020 are scored once on 2021–2024 as a terminal test. For each lead: mean-squared skill against climatology, raw persistence and damped persistence, anomaly correlation and amplitude ratio, per channel and per scope (trained and held-out longitudes separately), from three starts per held-out year, with block-bootstrap intervals over (year, start) [13]. The null ladder beside every read-out is climatology, persistence, damped persistence (a per-cell AR(1) fitted on training years), the linear inverse model in data space and in the rung’s embedding space [11, 12], an analogue (retrieval) forecast, and the previous rung; a learned model is reported against all of them. A configuration is compared to another on the dense field; the transport and index read-outs are downstream checks and cannot resolve a difference at ten effective starts. The winning configuration of any comparison is replicated with a second seed before it is written up, and a negative result is reported exactly like a positive one.

8.1 The comparisons, in order

(1) The hourglass codec against a snapshot codec. Both frozen, both feeding the same forecaster: the one-step forecast ratio and the future-cone families against their bars, and the T1–T2 gap inside the hourglass model, which is what history buys.

(2) The learnable geometry. Three arms on the same tensor, the same codec size, the same steps, three seeds each: **A0**, the fixed hourglass with the physics-set geometry — the control and the evaluation geometry for every arm; **A1**, the eight numbers per region and the per-channel scales and memories learned gate-then-harden from A0’s initialisation; **A2**, as A1 but with the ocean-surface drift primed at $-\mathbf{u}_{\text{clim}}\Delta t$, the climatological surface current reversed, which separates whether the gradient can find upstream on its own from whether it can only keep it. Five read-outs, written before any run: **R1**, the T1 future-cone loss per lead against its bar; **R2**, the T1–T2 gap; **R3**, the correlation of the learned surface drift with $-\mathbf{u}_{\text{clim}}$ over the world ocean where $|\mathbf{u}_{\text{clim}}| > 0.2 \text{ m s}^{-1}$, a clearly positive value meaning the cone learned “upstream” from the data alone; **R4**, the western-boundary depth check at an anchor just north of the 26.5°N array — the learned surface drift should point south-west, up the Florida Current, and the attention mass on deep ($> 1,000$ dbar) profile tokens should sit north of the anchor, up the deep western boundary current, with shallow ones south; **R5**, the target-variance flag. *Verdict rule:* A1 is adopted if it beats A0 on R1 at every lead ≤ 3 , paired at all three seeds, with R5 clean; if A2 beats A1 and A1 matches A0, the gradient cannot find upstream and the geometry is set from climatology rather than learned; if neither wins, A0 stays and the next design is a per-example cone whose drift is predicted from the waist, not a bigger static one. The first ablation after the arms is A1 without the far ring, which measures whether the ring pays for its tokens rather than arguing it.

(3) The forecaster. The linear backbone with a residual against the same transformer without it; direct multi-lead targets against one-step autoregression rolled; log-spaced against dense lags; each on the same codec, the same starts and the same battery.

(4) The footprint and the tiers. Coarse groups re-read as offset tokens with their true footprint against nearest-cell copies, on the same anchors and seed — no change in held-out per-family loss falsifies the footprint fields at this scale. The tier-P stores added to the cone against the cone without them, per read-out goal: El Niño through Niño-3.4 at a six-month lead against persistence of the warm-water volume; the carbon sink through held-out surface- CO_2 reconstruction against a climatological null; the overturning through the held-out per-family loss on velocity with drifter tokens against without; sea-surface temperature through the held-out loss at level-3 cells with drifter and ship tokens against without. The first tile codec, coastal, over a few hundred cells where colour, sea-surface temperature and profiles all have coverage, with and without the tile tokens, three seeds; parity falsifies the fine tier for

the pentad forecast and leaves it for fine targets. AlphaEarth’s embedding in place of the programme’s tile codec, everything else fixed, measures whether the tile codec is worth its compute.

(5) Scale. At L2 the width ladder is re-measured on the rolled field; a flat ladder means the rung is data-limited and the next data rung, not the next parameter rung, is bought. At L3 the sphere-routed mixture of experts against a dense model of the same active parameters.

Each comparison is one line in a configuration; none affects the build order, which is geometry, task mix, aperture gate, parameter maps and harden step, forecaster, tiers.

9 Results

This section is reserved. Nothing in this paper has been trained; every number that is a choice is marked as one above, and the tables below are the ones the comparisons of §8.1 will fill, with their columns fixed now so that the reading cannot be shaped by the result. Each cell will carry the seed count and, where a pair or triple exists, its spread.

codec	one-step ratio	lead 1 / bar	lead 3 / bar	lead 6 / bar	T1–T2 gap
snapshot (waist only)	—	—	—	—	—
hourglass, fixed geometry (A0)	—	—	—	—	—

Table 3: Comparison (1): the hourglass codec against a snapshot codec, both frozen, both feeding the same forecaster. Ratios are held-out loss over the predict-the-mean bar; the one-step ratio is the forecaster’s z -space error relative to persistence.

arm	R1 lead 1	R1 lead 2	R1 lead 3	R2	R3	R5
A0 fixed	—	—	—	—	0 by construction	—
A1 learned	—	—	—	—	—	—
A2 learned, primed	—	—	—	—	—	—
A1 without the far ring	—	—	—	—	—	—

Table 4: Comparison (2): the learnable geometry, three seeds per arm, paired by seed. R4 — the drift direction and the attention-by-depth at the western-boundary anchor — is a figure.

lead (pentads)	1	3	8	21	34	55	73
persistence, MSSS _{clim}	—	—	—	—	—	—	—
damped persistence	—	—	—	—	—	—	—
LIM, data space	—	—	—	—	—	—	—
LIM, embedding space	—	—	—	—	—	—	—
analogue	—	—	—	—	—	—	—
forecaster, linear backbone + residual	—	—	—	—	—	—	—
without the backbone	—	—	—	—	—	—	—
one-step, rolled	—	—	—	—	—	—	—
dense lags	—	—	—	—	—	—	—

Table 5: Comparison (3) and the null ladder: mean-squared skill against climatology on the dense field, world ocean scope, per lead; the same table is produced for MSSS_{damped}, anomaly correlation and amplitude ratio, per channel and per scope.

read-out	score	null	without the stores	with the stores
El Niño	Niño-3.4 correlation at a six-month lead	persistence of the warm-water volume	—	—
carbon sink	held-out surface-CO ₂ error	climatological null	—	—
overturning	held-out velocity loss / bar; transport r at 26.5°N	drifter-free cone; damped persistence	—	—
sea-surface temperature	held-out level-3 cell loss / bar	drifter- and ship-free cone	—	—

Table 6: Comparison (4): the point-observation stores per read-out goal, and the footprint fields, each against the same cone without them.

width	held-out minimum step	one-step ratio	mean ACC, leads 1–73	mean MSSS _{damped}
—	—	—	—	—

Table 7: Comparison (5): the width ladder re-measured at L2 on the rolled field, one row per width.

10 Related work

Learned forecasting of the atmosphere and ocean. Global data-driven weather models [15, 16, 17] and their hybrids with a dynamical core [18] forecast full reanalysis fields at daily leads; foundation models of the atmosphere [19, 20] extend the same architectures across variables and resolutions, and end-to-end systems that read observations rather than a reanalysis [21] are the closest in intent to the observational footing taken here. Learned ocean emulators [22, 23] and coupled atmosphere–ocean emulators [24] operate at monthly to seasonal leads on gridded reanalysis fields. All of them read one grid, one variable set and one reach; the design here differs in reading every source at its own footprint through one token schema, in sampling a per-channel cone rather than a fixed receptive field, and in predicting through a query that carries the footprint of the answer.

Speed times time as a receptive field. The argument that the relevant neighbourhood of a point is set by a signal speed and a horizon has precedents: an ocean emulator that drops lateral boundary conditions because eddies travel a bounded distance over the forecast horizon [23]; a neural semi-Lagrangian layer that gathers at $\mathbf{x} - \Delta t \mathbf{u}$ and turns a CFL condition into a Lipschitz bound [25]; limited-area models with fixed halos; and the deformable sampling of attention at learned offsets with a learning-rate multiplier on the offsets [4], from which the gate-then-harden schedule and the multiplier are taken. Truncation by memory, a union of cones per channel, and a learnable ellipse with the direction shared per flow and the scale per channel are, as far as the authors know, not stated as a principle elsewhere.

Representation learning for Earth observation. The architectural basis is the masked autoencoder [5] over tokens and Perceiver-style cross-attention into a fixed latent set for inputs of variable size and modality [6]. Geospatial foundation models for imagery [26] embed tiles; AlphaEarth Foundations [2] produces a per-pixel embedding field of the land from multi-sensor sequences, conditions a per-source decoder on sensor metadata, and is the closest statement of the embedding-field idea, over the land surface and at annual cadence; its published embedding enters this design as a token cache. The evaluation of a representation through frozen backbones with shallow read-outs [27] is the stance taken for the read-outs here, with the addition that the representation is judged by what a forecaster can predict from it rather than by a current-state probe.

Empirical nulls and verification. The linear inverse model [11] is the standard empirical forecast against which learned systems at these leads are judged, including as a benchmark for decadal forecasts [12]; the verification framework used here — skill scores against climatology and persistence, anomaly correlation per lead, a damped-persistence reference, bootstrap intervals — is the one established for interannual-to-decadal prediction [13]. The decoupled heteroscedastic objective follows [7]; the fair continuous ranked probability score [14] is the probabilistic score; the data-constrained scaling rule of at

most four epochs follows [8]; the sphere-routed mixture of experts follows the sparse-expert transformers of [10].

Observing the read-outs. The transport at 26.5°N has been measured continuously since 2004 [28], and the Argo array [29] is the interior’s observing system; both enter here as observations in their own coordinates, the arrays’ instruments as a tier-P store and the profiles as tokens, rather than as a target index alone.

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A Notation

symbol	meaning
anchor	the cell and pentad an embedding is computed for and a prediction is made from
pentad, bin b	the five-day averaging window $[5b, 5b + 5)$ days after 1982-01-01
lag ℓ , lead ℓ	pentads before and after the anchor
z	the stage-1 embedding of an anchor, width 128–256
$(\Delta x, \Delta y, \Delta z, \Delta t)$	offset of a token from the anchor in km, m and days
$(\log_2 f_p, \log_2 f_t)$	footprint: spatial support in 0.25°-cell units and temporal support in pentads, base-2 logarithms
$\mathbf{d}, \Sigma_0, \Sigma_v$	drift per pentad, waist ellipse, velocity-spread ellipse; $\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v$
$S = \log \Sigma$	the matrix logarithm in which an ellipse is stored
s_c, τ_c	per-channel log-scale and memory of the aperture
$w_{c,i,\ell}$	the aperture weight of point i at lag ℓ for channel c
v_{design}	1.5× the fastest measured signal of a family; 5.4 ms^{-1} for the ocean
$\mathbf{G}_\ell, f_\theta$	the contractive linear propagator and the residual transformer of stage 2
MSSS_x	$1 - \text{MSE}/\text{MSE}_x$ against reference x
ACC, a	anomaly correlation and amplitude ratio per lead
bar	the score of predicting the mean, per task, family and lead

B Data and code availability

Code: <https://github.com/blauewelt/earth> — the cone geometry (`ml/cone.py`), the sampler (`ml/cone_sampler.py`), the codec and its masking plan (`ml/cone_codec.py`, `ml/train_cone.py`), the point-store reader (`ml/family10_store.py`) and the store builders (`ml/build_family7.py`, `ml/build_family8_argo.py`, `ml/build_family10_stores.py`). Data: the global tensor (`family7_global1025_pentad_l2`), the Argo store (family 8), the point-and-track stores (families 10.1 and 10.2) and the fine-granularity families as they are built are published in the Hugging Face dataset `chfrank/earth-tensors`, each with a registry listing every file’s checksum, sources and licence; stores whose licences forbid redistribution are listed there and held in a private repository. The design notes for the families are under `ml/paper/notes/`; the experiment plans and their verdict rules under `ml/plans/`.

Sources: GLORYS12 and the GREP ensemble (E.U. Copernicus Marine Service Information); OISST v2.1 (NOAA NCEI); NCEP/NCAR Reanalysis-1 (NOAA PSL); the Roemmich–Gilson Argo climatology (Scripps; Argo data are collected and made freely available by the International Argo Program and the national programs that contribute to it); OC-CCI v6.0 (ESA Climate Change Initiative, PML); the RAPID-MOCHA-WBTS transport time series (NERC, NSF, NOAA); the Global Drifter Program and the Global Tropical Moored Buoy Array (NOAA AOML, PMEL); SOCAT; the Copernicus Marine along-track sea-level products; and the sources named per store in the family registries.